Unit 11, Lesson 8: Barbie Bungee – Part 2

Benchmarks

MA.8.DP.1.1 Given a set of real-world bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

MA.8.DP.1.3 Given a scatter plot with a linear association, informally fit a straight line.

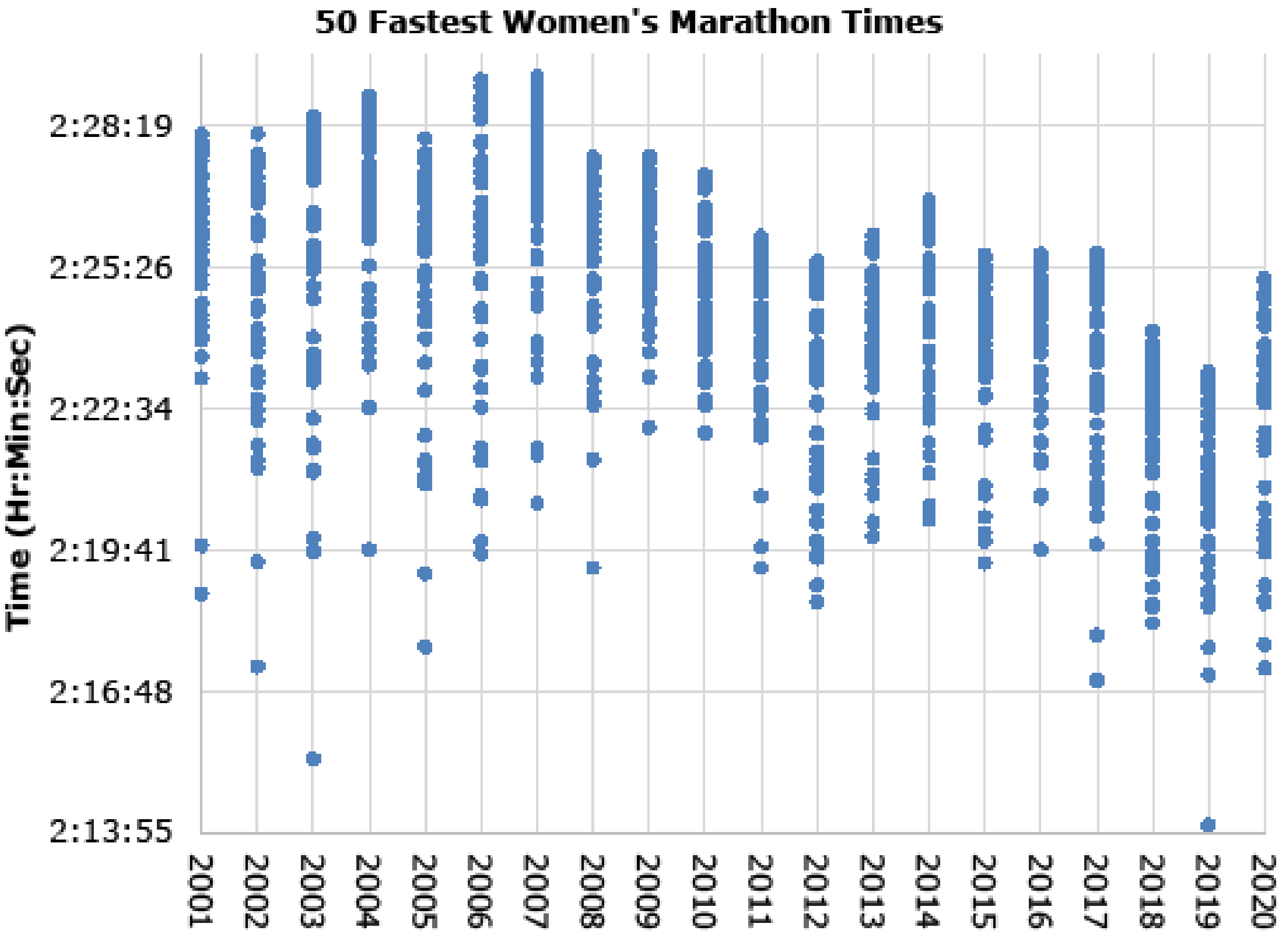
MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

Learning Targets

- ☐ **I can collect data for an experiment.**
- ☐ **I can construct a scatter plot for a set of bivariate data.**
- ☐ **I can describe patterns of associations in data on a scatter plot.**
- ☐ **I can informally fit a straight line to data on a scatter plot.**
- ☐ **I can interpret the slope and y-intercept of a linear equation in a real-world context.**

11.8.1 Warm-Up: What's Going on in the Graph?

1. The graph shows the times of the 50 fastest women's marathoners in the world from 2001 to 2020.



a. Describe how you know if the runners are getting faster or slower each year.

b. Give at least one possible explanation for the trend.

11.8.2 Activity: Barbie Bungee – Part 2

In the previous lesson, you performed an experiment and recorded the data for Barbie's bungee experience. Work with your group to represent and analyze the data collected.

Representing and Analyzing the Data

1. Construct a scatter plot to represent the data collected, using the average of the trials for each number of rubber bands.

    - Label the axes with the appropriate variables.
    - Identify and label an appropriate scale for each axis on the grid.
    - Plot a point for each pair of values in the data set.
    - Title the graph.

2. Interpret the data represented on your scatter plot.

a. Describe the relationship between the number of rubber bands used and the jump distance.

b. Describe any patterns of association in the scatter plot.

c. Informally fit a line to the data using a straightedge.

3. Discuss the following questions with your partner.

a. How do you think the data would change if Barbie were taller?

b. How do you think the data would change if Barbie were lighter?

11.8.3 Bring It Together: Interpreting Data

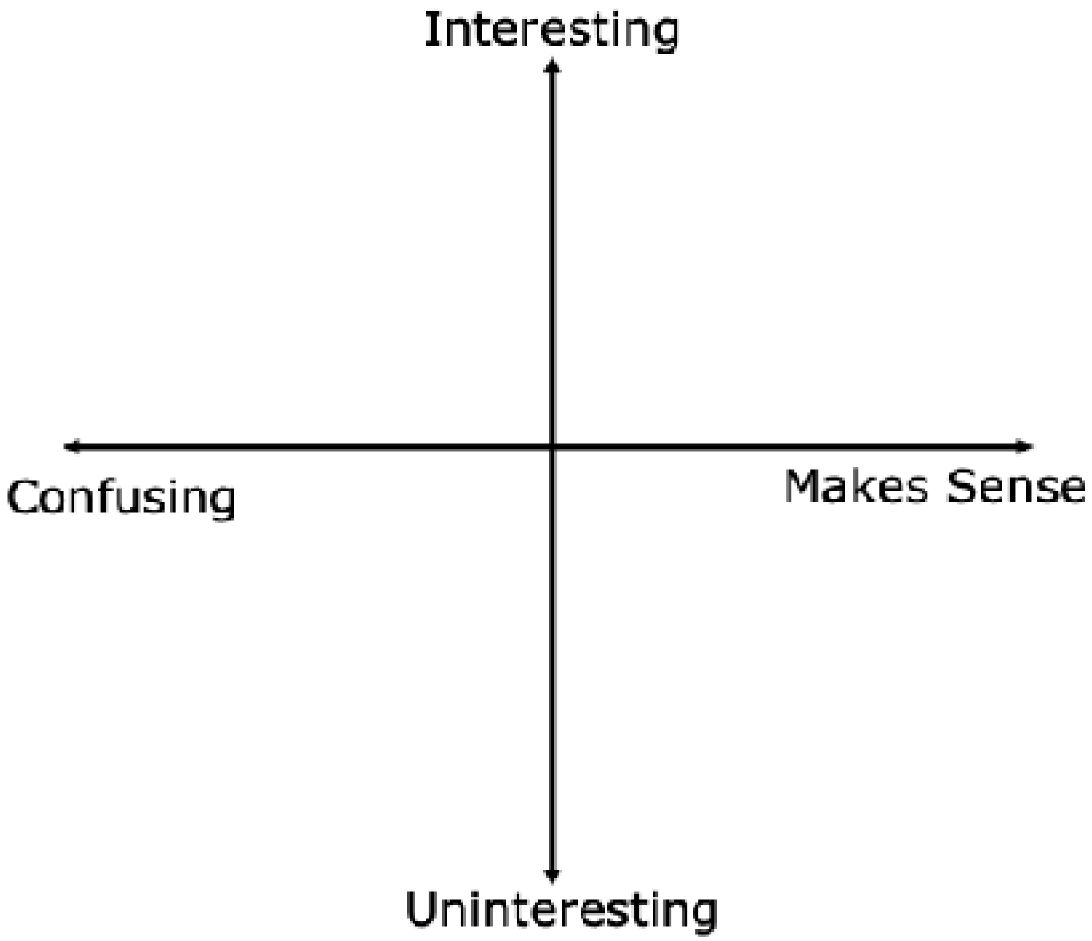
1. Noah performed the same Barbie Bungee experiment. He fit the line $y = 7.2x + 22.17$ to the data in his scatter plot, where x represents the number of rubber bands used and y represents the jump distance in centimeters.

   a. Interpret the slope of the line fit to Noah's data.

   b. Discuss with your partner whether or not you think Noah's line could be a good fit for the data you collected in the experiment.

11.8.4 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your thoughts about the Barbie Bungee activity.



2. **Explain why you placed your point there.**